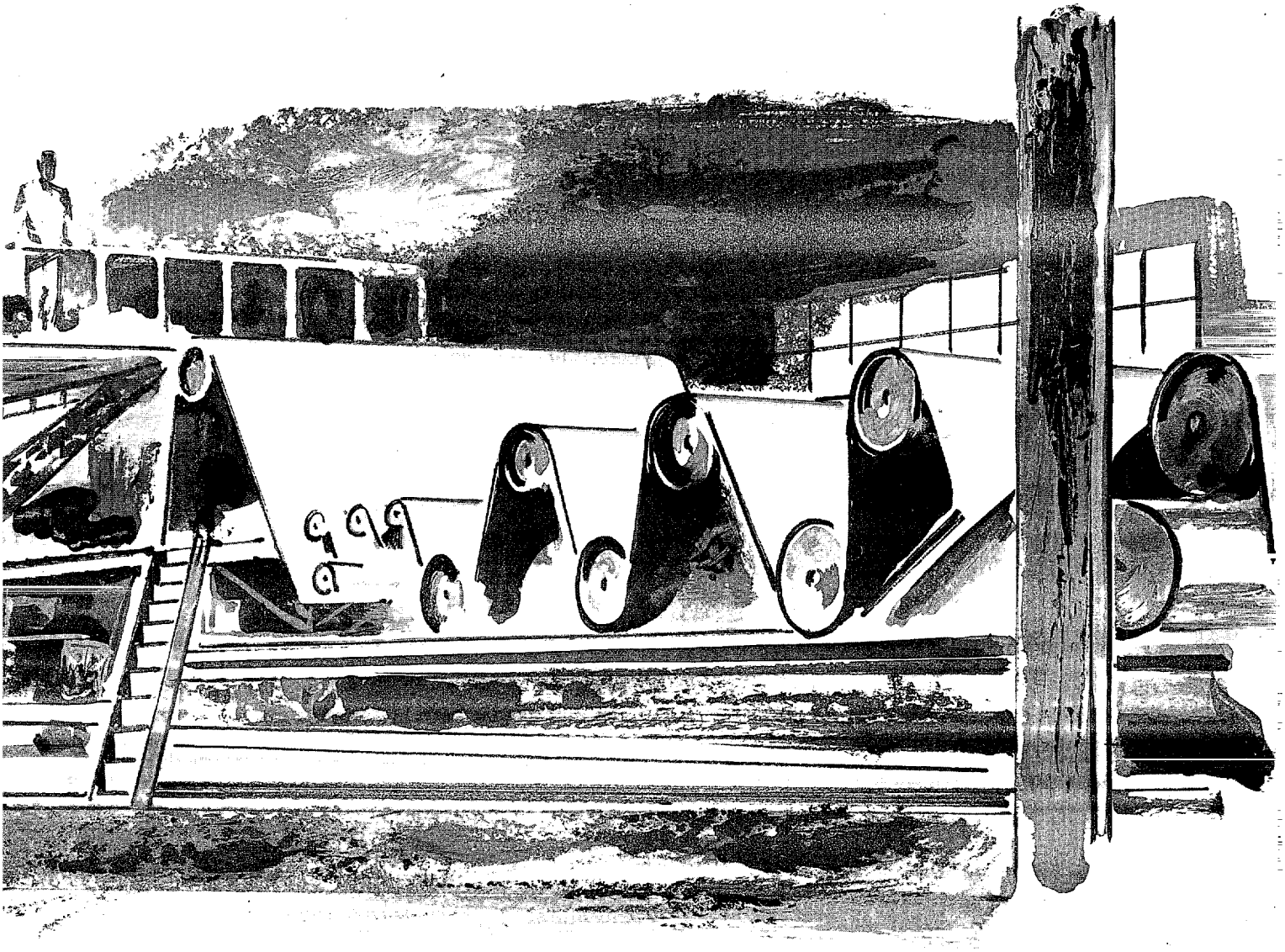
INSTRUCTIONS

GEK-7876

# REGENERATIVE BRAKING WINDER CONTROL

## FOR DRIVES WITH SILCOMATIC* TYPE POWER SUPPLY



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GENERAL ELECTRIC

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*These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.*

# REGENERATIVE BRAKING WINDER CONTROL FOR DRIVES WITH SILCOMATIC TYPE POWER SUPPLY

## INTRODUCTION

These instructions describe the operation, adjustments, and maintenance of the electrical control equipment for a regenerative brake winder used in paper mill applications. It is assumed that the reader is familiar with the mechanical equipment and general operation of paper mill winders.

The description, illustrations, and circuits used in these instructions are for a typical control. The elementary diagram and other drawings furnished with a specific equipment will show and explain the optional or additional features for that equipment.

Figure 1A shows, in simplified circuit form, how the basic electrical equipment is connected together as a system. The windup motor speed (and therefore sheet speed) is regulated by controlling the Silcomatic voltage. This is done by means of the windup regulator amplifier which varies the control signal to the Silcomatic. The unwind motor torque is controlled by regulating the unwind motor armature current and the unwind motor counter EMF. The unwind motor current is regulated to a value proportional to the sheet tension desired by means of the unwind regulator amplifier which varies the excitation to the field of the unwind booster generator. For a fixed unwind motor field excitation,

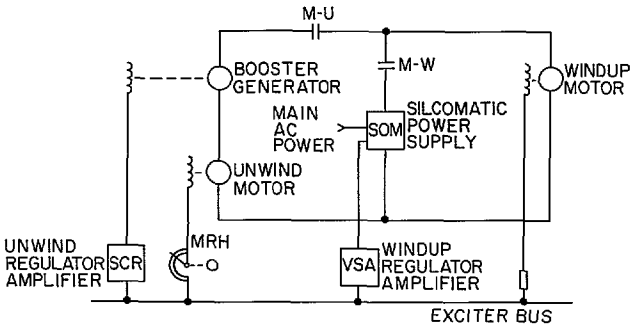



*Fig. 1A. Simplified diagram of basic elements of a regenerative brake winder drive*

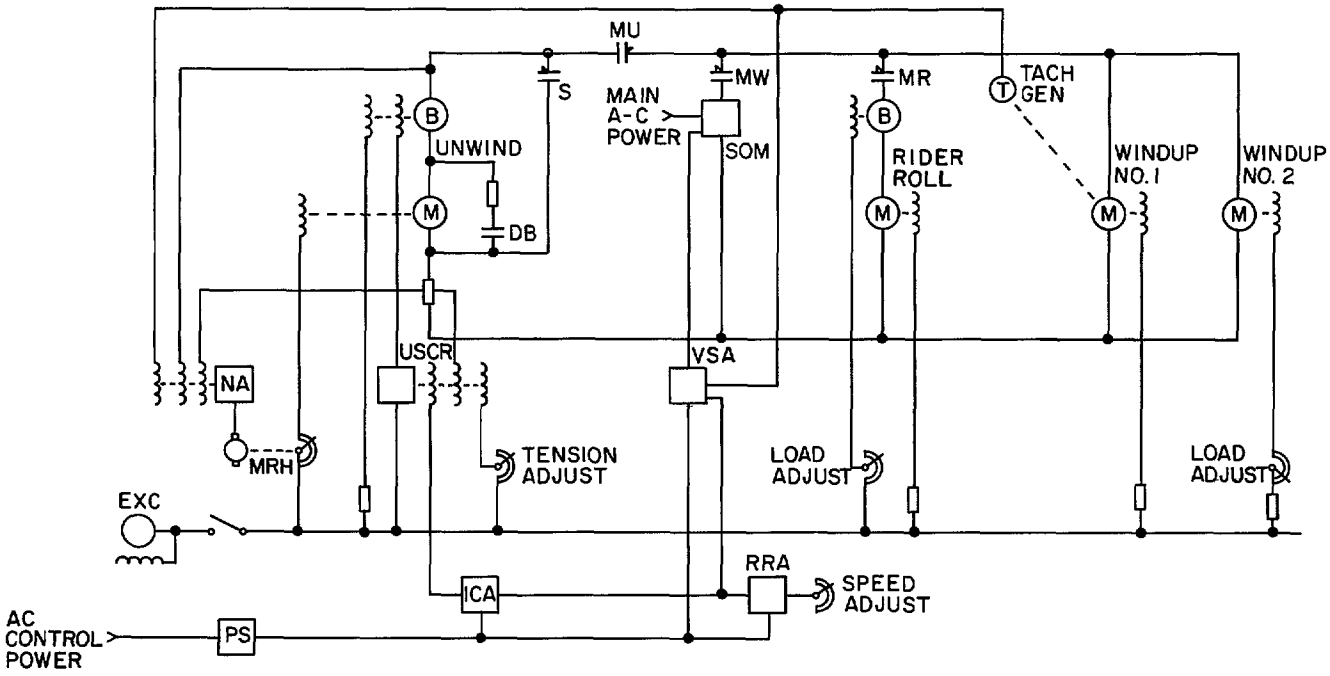



*Fig. 1B. Simplified diagram of typical regenerative brake winder drive*

regulating the armature current will regulate the torque of the motor and therefore (assuming no friction losses and constant roll diameter) the sheet tension. However, the roll diameter constantly decreases as the paper is unwound. To accommodate this fact, the unwind motor field strength is changed in proportion to the unwind roll diameter. This is done by means of the unwind CEMF regulator, which acts through a motor-operated rheostat to weaken the unwind motor field strength in proportion to the roll diameter builddown.

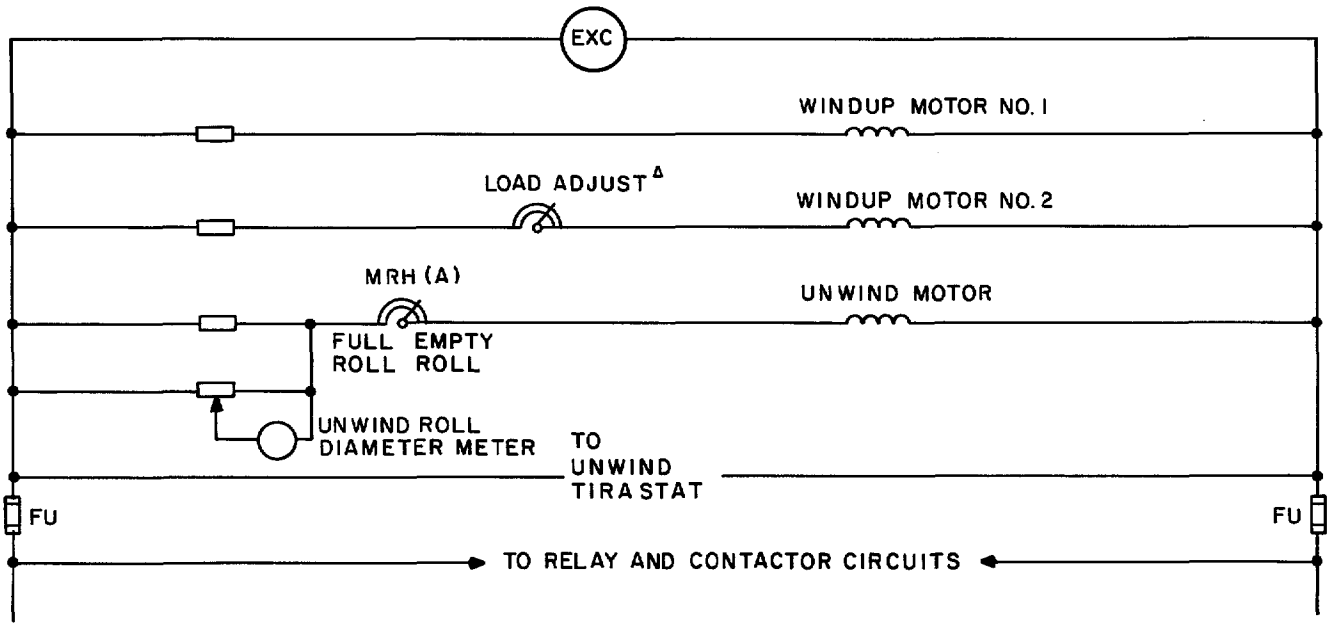
Figure 1B shows, in simplified circuit form, more of the equipment associated with a typical regenerative brake winder drive. The regulator modules are identified as follows:

| | |
|---|---|
| VSA | Transistorized amplifier for control of windup speed |
| RRA | Rate reference unit to provide timed acceleration and deceleration |
| PS | Power supply for transistorized amplifier |
| SCR | Controlled rectifier regulator-exciter for control of unwind current |
| NA | Magnetic amplifier used as part of unwind CEMF regulator |
| MRH | Motor-operated rheostat used as part of unwind CEMF regulator |
| ICA | Inertia compensation amplifier |

A typical rider roll drive is shown. Slitters are also usually provided; the equipment elementary diagram shows details of this feature.

## MAGNETIC CONTROL CIRCUITS

Figures 2A and 2B show the magnetic control circuits for a typical regenerative brake winder control.



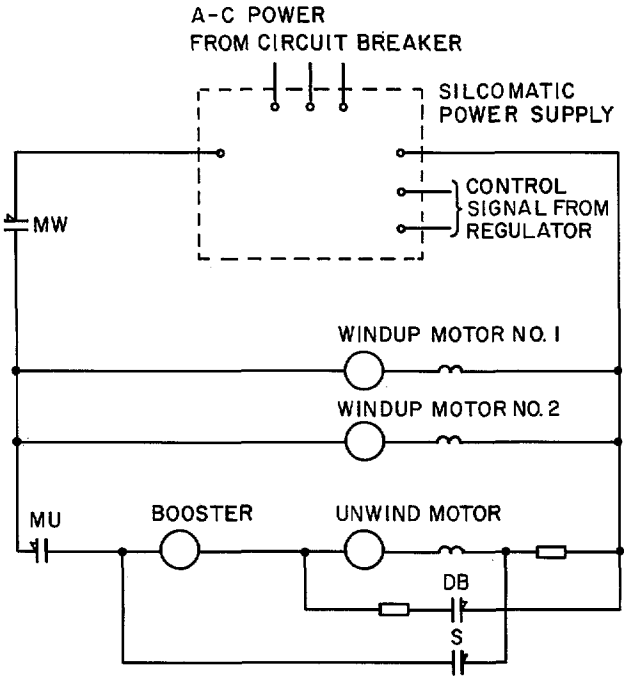




*Fig. 2A. Typical magnetic control circuit (see Fig. 2B)*

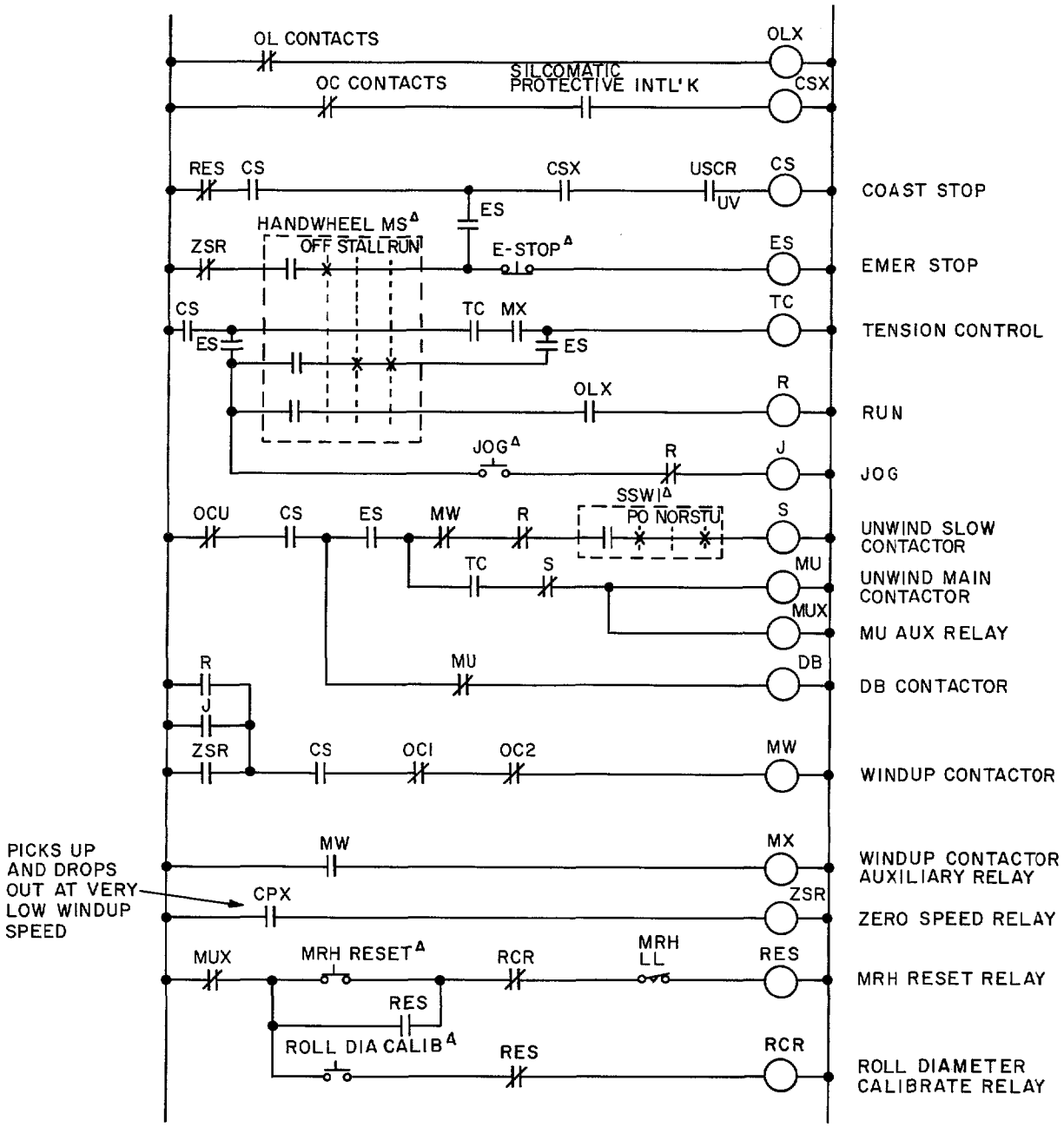



*Fig. 2B. Typical magnetic control circuit (cont.)*

The functions of the various circuits and relays are listed beside the relay coils. Note that in most cases the relay designation is an abbreviation of the relay function. The operator's Handwheel Controller RH is used to start, stop, and control the speed of the windup and to apply stall tension or run tension to the unwind. This device also has a rheostat connected to it that varies the speed reference signal to the windup speed regulator. The operator's Selector Switch SSW1 is used to initiate payout or slack take-up operation of the unwind. Notice that this switch energizes the unwind slow contactor S, which connects the unwind motor to the booster only. Other contacts on SSW1 signal the booster regulator to provide positive or negative voltage as required to operate the unwind motor in the payout or slack take-up mode. Contacts on the MRH Reset Relay RES and the Roll Diameter Calibrate Relay RCR are used to energize the Motor-operated Rheostat (MRH) that controls the unwind motor field. This permits

setting the MRH to the proper position before starting the machine after a new roll has been put into the unwind stand. The contact labeled MRH-LL is the end of the travel limit switch on the motor-operated rheostat.

The interlock labeled CPX (in the ZSR coil circuit) is an interlock on a buffer a-c relay. The speed detecting relay CP, located in the module with the regulator power supply, closes when the speed (tach volts) signal is above a few percent of rated speed and opens when the speed drops to nearly zero speed.

Most drives will also include optional features which add to the basic circuits shown here. For details of these features, refer to the elementary diagram of the specific drive in question.

## REGULATOR

### GENERAL

The regenerative brake winder control contains three separate regulators:

- Windup speed regulator
- Unwind current regulator
- Unwind motor CEMF regulator

The over-all function of each of these regulators was pointed out in the Introduction. The following is a more detailed description of their operation.

### WINDUP SPEED REGULATOR

Figure 3 shows a simplified diagram of the basic windup regulator. Most of the elements of the regulator circuits are physically contained in the following modules:

PS5 Power supply module (see instruction book GEK-7906)

RRA Rate reference module (see instruction book GEK-7905)

VSA Voltage and speed amplifier module (see instruction book GEK-7907)

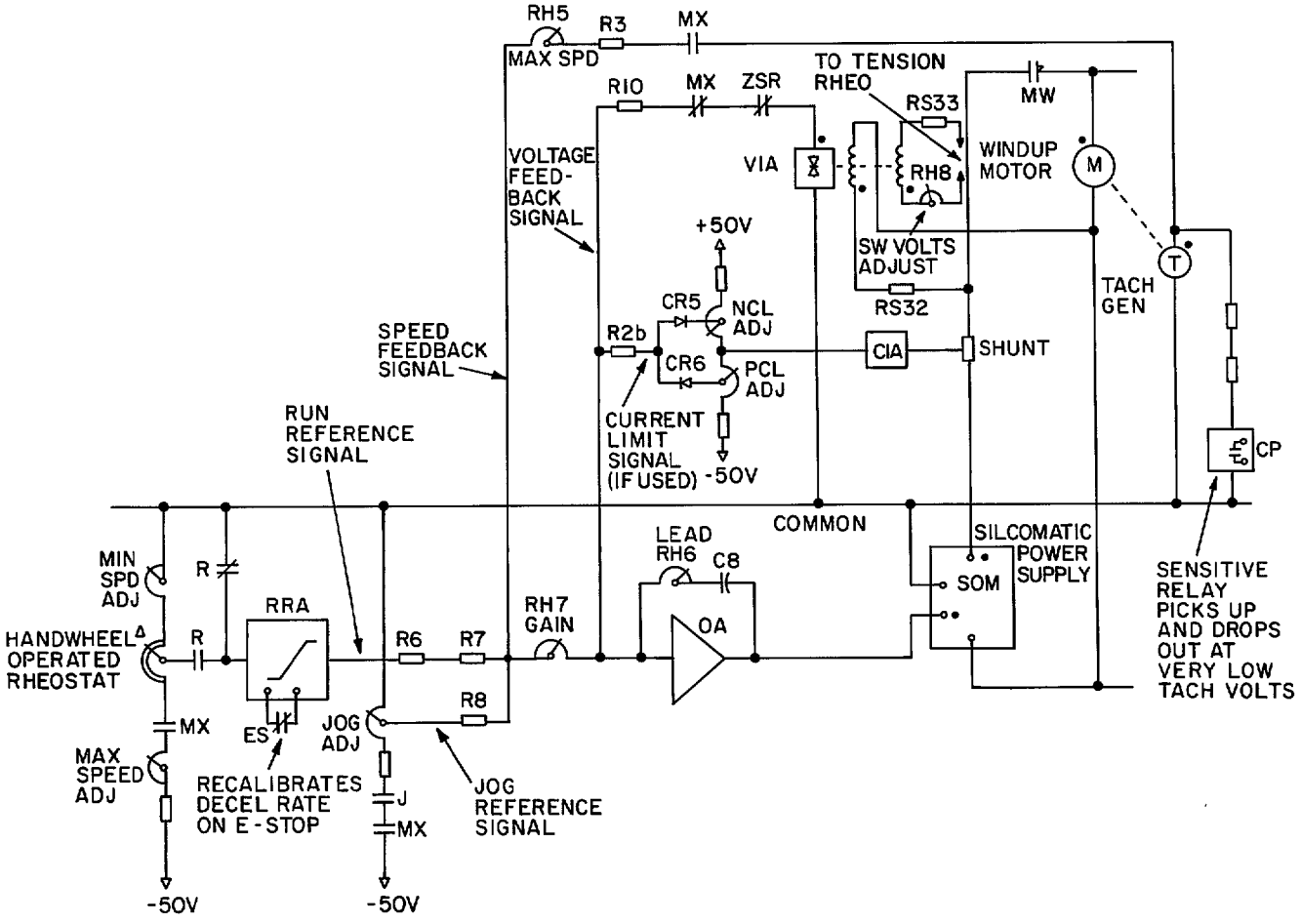



*Fig. 3. Simplified circuit of basic windup regulator*

The instruction books describe the operation of the modules. However, the following brief explanation of the over-all regulator action will help to tie the various pieces together.

There are four significant modes of operation for the system:

- Standstill (when windup contactor is open)
- Normal operation
- Emergency stop operation
- Jog operation

### Standstill

When the drive is at standstill, the run relay R, the jog relay J, the windup contactor MW, and the contactor auxiliary relay MX are dropped out. In this mode the run reference signal and the jog reference signal are zero and can be ignored. However, in the voltage isolation amplifier VIA, the tension reference signal from the unwind regulator (through resistor RS33) is compared to the Silcomatic terminal voltage (via resistor RS32). Any difference between these two voltages results in a voltage on the output of VIA. This voltage is fed to the operational amplifier via resistor R10. The polarities of the various elements are such that the VIA output signals the operational amplifier to apply a voltage to the Silcomatic input that causes the Silcomatic voltage to approach the motor voltage. The operational amplifier gain is extremely high - approximately 0.006 microamp input current produces full output of 25 volts – so the Silcomatic voltage is maintained equal to the motor voltage. This is done to minimize sheet tension transients at the instant the drive is transferred from stall tension to the run condition, since the unwind booster, when in stall tension, causes a current to flow in the windup motors, thereby causing a voltage drop across the line contactor M. The level of Silcomatic voltage is set by the level of the tension rheostat and the switch voltage adjust rheostat RH8. RH8 is set so that Silcomatic voltage does not change when the zero speed relay drops out after a drive stop initiation. Thus, the energy of any paper roll motors will be dissipated through the Silcomatic power supply. If the unwind regulator is not selected, then the signal to RS33 will be zero and the Silcomatic voltage will be the only excitation to VIA. If any voltage exists, VIA will cause OA to drive the Silcomatic to zero volts, thereby suiciding the Silcomatic voltage to zero.

### Normal Operation

When the drive is energized in the normal (run) mode, the run relay R, the windup contactor MW, and the contactor auxiliary relay MX are energized. A portion of the voltage across the handwheel-operated rheostat is applied to the input of the rate reference amplifier module RRA. This module contains a transistorized ramp generator whose output voltage gradually increases (or decreases) to the level of the input voltage. (See the RRA instruction book GEK-7905 for more complete information on this module.) The output of the rate reference amplifier RRA is applied to the operational amplifier OA input via resistors R6 and R7 as a run reference signal. This signal causes the operational amplifier output voltage to increase, the Silcomatic voltage to increase, and the motor to rotate. The motor rotation causes the tachometer generator to produce a voltage and this voltage is fed back to the operational amplifier via the normally open MX contact and the maximum speed pot RH5 – the polarity of RH5 – tends to reduce the operational amplifier and Silcomatic output voltages. When the feedback signal strength matches the reference signal strength the regulator is balanced and the motor speed remains at that level.

If the operator moves the handwheel-operated rheostat to a higher setting, the input to the rate reference amplifier RRA is increased, its output gradually increases to that value, the reference signal gradually increases, and the operational amplifier output, the Silcomatic output, and the motor speed increase until the feedback signal again matches the reference signal. Thus the motor speed is regulated to a value proportional to the setting of the operator's handwheel-operated rheostat.

When the handwheel is turned to its off position, the run relay R drops out and zero voltage is applied to the input of the rate reference amplifier RRA. The RRA output gradually goes to zero, and the Silcomatic voltage and motor speed follow the reference signal down. When the motor speed (and therefore tach voltage) gets to a very low value, the speed relay CP drops out and de-energizes the line contactor MW, and the drive is returned to the Standstill mode.

### Emergency Stop Operation

When an emergency stop is initiated, the emergency stop relay ES and the run relay R drop out. An interlock on the ES relay recalibrates the deceleration rate circuit in the rate reference unit RRA to provide a fast stop. The motor speed continues to be regulated and follows the rapidly decreasing reference signal since the run relay is de-energized. When the motor speed is very low the speed relay CP drops out (as in a normal stop), the windup contactor is opened, and the drive returns to the Standstill mode.

### Jog Operation

When a jog is initiated (by pressing the JOG push-button) the jog relay J, the windup contactor M, and the contactor auxiliary relay MX pick up. A jog reference signal is applied via resistor R8 to the operational amplifier. The operation is now the same as for normal operation except that the jog reference signal is used instead of the normal run reference signal.

### Antihunt Feature

Without an antihunt feature the drive would be unstable – that is, it would oscillate. For antihunt purposes the gain rheostat RH7 plus the lead rheostat RH6 and capacitor C8 are provided. The circuit with capacitor C8 senses when the operational amplifier output is changing and feeds a signal back into the input of the operational amplifier in a direction to oppose the change. This action slows down the rate at which the operational amplifier output voltage will change. This prevents oscillation and thus stabilizes the drive. The gain rheostat RH7 adjusts the maximum speed error signal strength into the operational amplifier and so assists the capacitor C8 to control the rate at which the operational amplifier output voltage can change.

## UNWIND CURRENT REGULATOR

The circuit for a typical unwind current regulator is shown in Fig. 4. The major element in this circuit is the Tirastat, a power amplifier that uses controlled rectifiers to convert constant potential 250 VDC from the exciter bus to an adjustable d-c voltage. Its output is proportional to the sum of the signals on its various input windings, each signal being proportional to the number of turns in the winding times the current flowing in the winding. To avoid confusion it helps to identify the signal strength on each winding in terms of ampere turns. Thus if one winding has a positive 0.82 ampere-turn signal on it and another winding has a negative 0.78 ampere turn on it, the sum of the signals (called the net ampere turns) is 0.04 ampere turn. In this case the SCR output voltage would be the value that corresponds to 0.04 ampere turn.

As shown in Fig. 4, the booster has two fields. This arrangement is used because the Tirastat voltage cannot be reversed. If it were to supply the only excitation to the booster, the booster output voltage could not be regulated to zero or a negative value. The auxiliary field is excited to provide a constant negative flux in the booster. With this excitation alone the booster would produce a moderate voltage with a negative polarity. The main field is excited by the Tirastat to provide a controllable positive flux in the generator.

### Current Regulator

When the regulator is energized a current reference signal is applied to winding F5-F6. This causes the SCR output to increase, the booster voltage to increase, and armature current to flow. The armature current is in a direction to cause holdback action of the unwind motor. The armature current causes a voltage drop across the unwind motor commutator field plus armature resistor. This voltage is applied to the current feedback winding F10-F9 so that the Tirastat output tends to decrease. When the negative ampere turns provided by the feedback winding almost equal the positive ampere turns provided by the reference winding, the Tirastat and booster output no longer increase and the system remains at that level. If for any reason the armature current starts to change from this balanced condition, the feedback signal immediately changes and the net ampere turns in the Tirastat change. This change in net ampere turns changes the output of the Tirastat – and therefore the booster voltage – until the original value of armature current is restored. In other words, the armature current is regulated to a value proportional to the current reference signal.

### Stall Tension

When the drive is operated in stall tension the current regulator is energized but the windup contactor is not energized. In this condition the booster voltage circulates armature current in the unwind and windup motors (Fig. 1), causing holdback torque on the unwind and motoring (pulling) torque on the windup, thus keeping a tension in the sheet. To prevent damage to the motor commutators the current should be considerably below the motor rated current. For this purpose the current reference signal is reduced. This is done by the normally open MX interlock which inserts the stall tension rheostat in the current reference circuit.

### Inertia Compensation

To accelerate or decelerate the unwind stand without affecting the sheet tension, it is necessary to compensate for the inertia of the machine and the roll of paper. This is accomplished by signalling the current regulator to increase its holdback current during deceleration periods and to decrease its holdback current (or, if necessary, transfer to motoring current) during acceleration periods. This signal is provided by the inertia compensation circuit which is shown at the bottom of Fig. 4. Whenever the speed reference signal changes, a charging current passes through capacitor C6. This current is amplified by the transistorized operational amplifier OPNL AMPL and is applied to the inertia compensation winding F1-F2 of the Tirastat. This

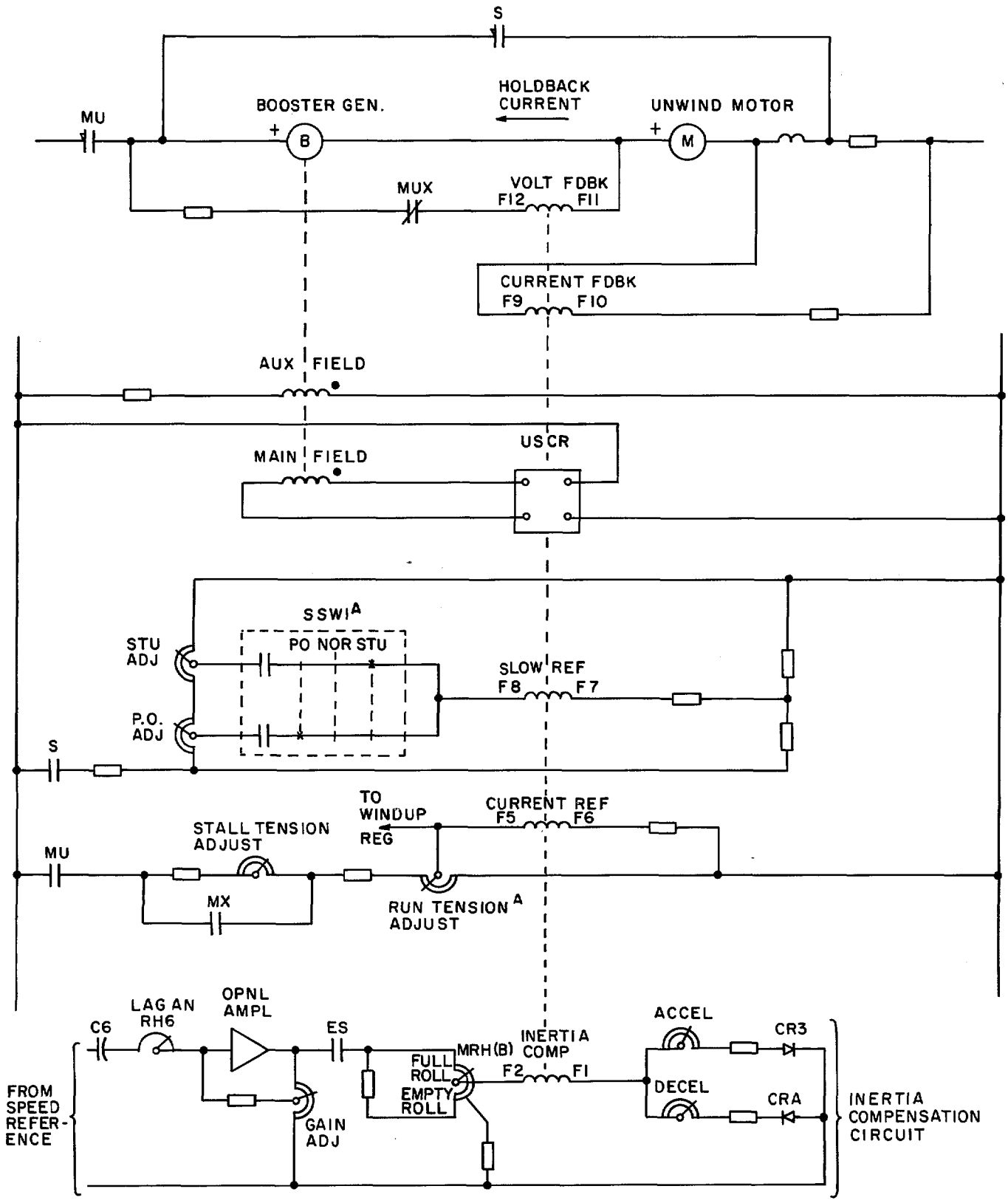



*Fig. 4. Typical unwind current regulator*

signal adds to (or subtracts from) the normal current reference signal in the magnetic flux of the Tirastat circuit. When the unwind roll diameter decreases its inertia decreases, so the signal fed to the inertia compensation winding must be tailored accordingly. This is done by the motor-operated rheostat MRH plate B. (The position of the MRH slider is proportional to the roll diameter, as noted in the discussion under the CEMF Regulator.) In some cases the inertia compensation signal required may be different for deceleration than for acceleration, even though the rate of speed change is the same. Detecting which condition exists and adjusting for it are done with diodes CR3 and CR4 and the ACCEL and DECEL adjust rheostats.

### Suicide

When the unwind line contactor MU is open (and relay MX is dropped out) the booster voltage is applied to a voltage feedback winding F11-F12. The polarity of the winding is such that for a positive booster voltage the Tirastat tends to turn off and therefore reduce the booster voltage. When the drive is stopped there is no reference signal applied to the Tirastat so this circuit holds the booster to essentially zero volts.

### Slack Take-up and Payout

When the STU-NOR-PO Selector Switch SSW1 is turned from its normal position to either the STU or the PO positions, the slow contactor S is energized, connecting the unwind motor directly across the booster. A slow reference signal is also applied to Tirastat winding F7-F8. Since the voltage feedback winding F11-F12 is still connected to the booster under this condition, the booster voltage is regulated to a value proportional to the slow reference signal. This permits motoring the unwind forward for sheet payout purposes or reverse for slack take-up purposes while the remainder of the winder is at rest.

## UNWIND CEMF REGULATOR

For proper operation of the system it is necessary for the unwind motor CEMF to be approximately proportional to sheet speed regardless of unwind roll diameter. As the unwind roll diameter decreases, the unwind RPM increases even though the sheet speed has not changed. To hold the CEMF constant as the RPM increases, the unwind motor field strength must be reduced as the roll diameter decreases. For this purpose an unwind motor CEMF regulator is supplied.

The circuit for a typical unwind CEMF regulator is shown in Fig. 5. The major element in this circuit is the magnetic amplifier (Amplistat). It is a power amplifier that converts a-c power to an adjustable d-c voltage. Its output is proportional to the sum of the signals on its various input windings, each signal being proportional to the number of turns in the winding times the current flowing in the winding. See also Appendix D for information on the Amplistat.

### CEMF Regulator

When the drive is started the tachometer generator voltage applies a signal to Amplistat winding F4-F3 in a direction to turn the Amplistat off. This must be compared to the CEMF of the motor. To get a signal proportional to CEMF the motor armature voltage minus the internal IR drop of the armature is considered. So a signal proportioned to the unwind motor armature voltage is applied to Amplistat winding F1-F2, and a signal proportional to the internal drop of the armature is applied to winding F6-F5. The polarities of these signals is such that:

Net signal = (motor voltage signal + IR drop signal) - tach volts signal

When this net signal is positive the Amplistat voltage rises above the minimum voltage set by the drop across resistor RS47 and current flows from the Amplistat through diode CR2, resistor RS47, and the armature of the MRH motor. This turns the MRH and reduces the unwind motor field strength, which in turn reduces the unwind motor armature voltage. When the armature voltage has fallen enough to reduce the Amplistat voltage to a value lower than the voltage drop across RS47, the MRH stops turning and a new equilibrium condition is established. As the roll diameter decreases further, the unwind motor rpm and armature voltage will increase and again cause the Amplistat voltage to rise, and again the MRH will turn or notch. This notching process continues as the unwind roll decreases in diameter.

A limit switch (MRH-HL) on the MRH prevents the regulator from trying to drive the MRH beyond its weak field travel limit.

Run and speed relay (R and ZSR) contacts prevent the regulator from trying to drive the MRH until the drive speed is at a reasonable value. This guards against slight misadjustments of the regulator causing the MRH to move when the drive is at standstill.

### MRH Reset and Roll Diameter Calibrate

The regulator can drive the MRH in one direction only. It can weaken the unwind motor field as the roll diameter decreases. To get the MRH back to the full roll (full field) position, the operator must

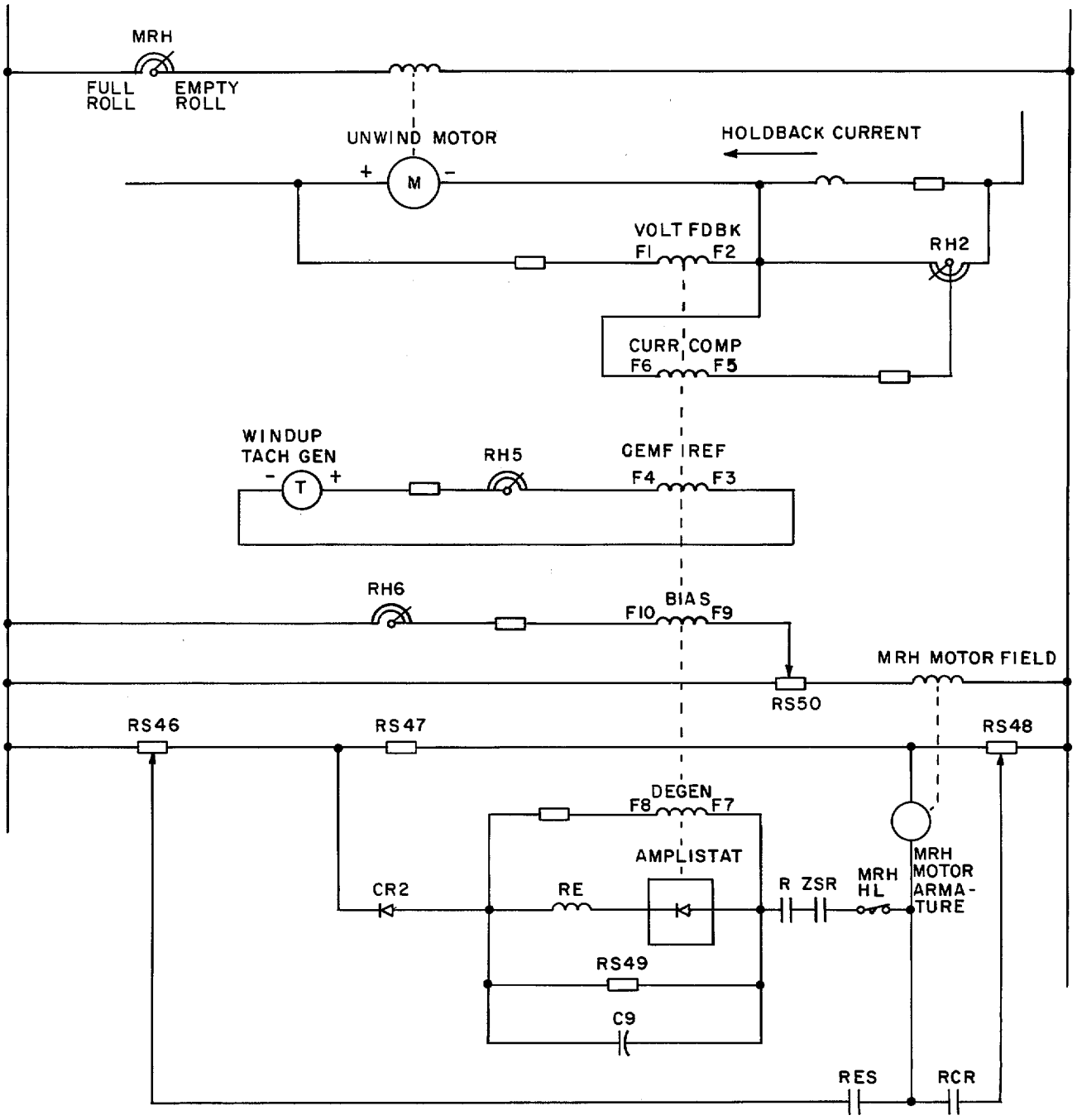



*Fig. 5. Typical unwind motor CEMF regulator*

press an MRH RESET pushbutton. This picks up relay RES, an interlock of which energizes the MRH motor to drive it back to the full roll position. If a roll smaller than full diameter is put into the unwind stand, or if the roll diameter is manually reduced by stripping some paper off the unwind roll, the MRH can be jogged around to the proper position for the existing roll diameter by pressing the Roll Diameter Calibrate pushbutton. This picks up relay RCR and energizes the MRH motor to drive it in the decrease field strength direction.

### Load Cell Tension Regulator (Optional)

In some cases a tension regulator that uses a hydraulic load cell to sense sheet tension may be supplied. The circuit used for this regulator is

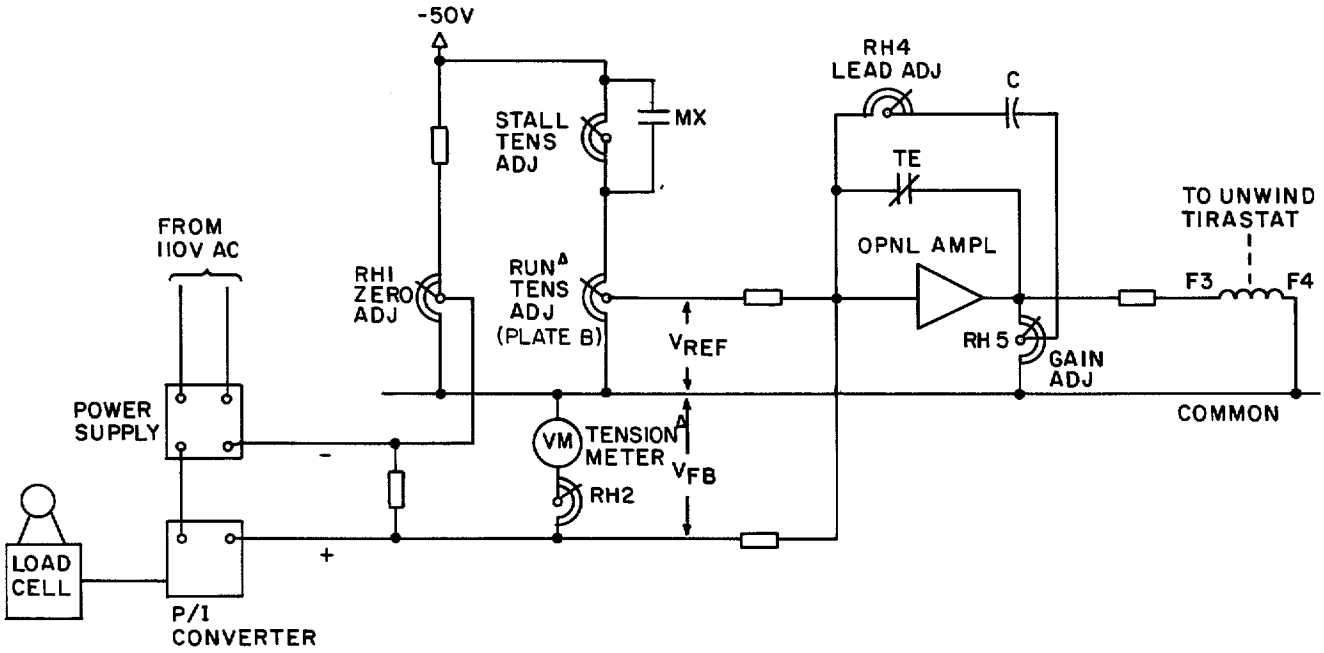



*Fig. 6. Load cell type tension regulator*

shown in simplified form in Fig. 6. The hydraulic load cell and pressure to current transducer (P/I converter) are mounted remote from the control panel.

The tension reference signal is received from plate B of the operator's Run Tension Adjust potentiometer (plate A of which is used for the reference to the unwind current regulator). The tension feedback signal is received from the load cell via the pressure to current transducer. The reference (V ref) is compared to the feedback (V fb) and the difference between the two is amplified by a transistorized operational amplifier (OPNL AMPL). The amplified signal is applied to the unwind Tirastat winding F3-F4 and acts as the reference signal to the current regulator (in place of the normal current reference signal on Tirastat winding F5-F6). This signal increases or decreases the unwind motor current as required to maintain the load cell (feedback) signal equal to the tension reference signal. Thus actual sheet tension is regulated.

The P/I transducer output will be from 9 to 11 milliamps when the actual sheet tension is zero. To compensate for this, a zero adjustment pot RH1 is provided in the tension amplifier module.

The gain of the tension amplifier can be changed by changing the value of capacitor C or by adjusting the gain pot RH5.

A "lead" signal is provided to compensate for the inherent lags in the tension sensing units. Once the value of capacitor C has been set, this lead effect can be adjusted by the lead adjust rheostat RH4.

Reduction in tension for stall conditions is accomplished in the same manner here as for the current regulator - that is, when relay MX drops out the tension reference is reduced.

## INSTALLATION AND TUNE-UP

### PRELIMINARY INSPECTION

1. Upon receipt of the equipment, it should be uncrated at once and carefully inspected for damage incurred during shipment.

2. If the equipment is not to be installed immediately it should be stored in a clean, dry location.

3. After the equipment has been uncrated and installed, it should be thoroughly cleaned with a soft brush, cloth or low-pressure, dry compressed air.

4. Manually operate all devices that have moving parts to assure that they are free from binding or misalignment. Remove all temporary shipping seals, locks and bracings.

5. Check to assure that all interconnecting wires and connections to remote located devices are in agreement with the elementary diagram furnished with the equipment.

6. Check all shielded wires to insure that terminations have been made in accordance with the interconnection diagram.

7. Check all terminals for loose connections, adequate electrical clearance, etc.

8. Inspect all contact-making devices to insure that the contacts are clean and properly positioned.

9. Check generator fields for proper resistance as shown on the machine data sheet of the elementary diagram. Check also for grounds in machine field circuits, being careful to avoid subjecting the control equipment to voltages which might damage semiconductor components.

10. Insure that the various voltage and current feedback signals are being extracted from the proper points on the Silcomatic, booster, and motors.

11. Insure that adequate electrical, mechanical and thermal clearances have been maintained.

12. Establish a method of adequate communication between the control panels, the operator's stations and the machinery before any attempt is made to start and tune up the drive.

## TUNE-UP PROCEDURE

1. General

Experience has shown that following certain prescribed procedures will result in a more trouble-free, faster and less expensive installation. Some of these procedures are listed below. These instructions should not be interpreted to cover all conditions which might be encountered or need to be considered during the tune-up period. The actual start-up and tune-up procedure should be established beforehand by those responsible, giving due regard to the safety of personnel and the protection of the electrical and mechanical equipment involved. The procedures outlined here assume that the rotating machinery and other remote mounted equipment are all in proper working order. It is also assumed that the Preliminary Inspection has been completed and the electrical system is free from grounds, haywires, etc.

2. Initial Conditions

Before applying any power to the control equipment, first open the control power switches and disconnect one motor armature circuit cable to each motor at the control panel. Also disconnect one end of each booster and motor field, one side of all Tirastat excitation (F) control windings except the bias winding, and one end of each Amplistat control winding except the bias winding.

3. Exciter Voltage

Start the MG set. Adjust the resistor in series with the exciter shunt field for 250 VDC output. After the exciter has reached its normal operating temperature, readjust this resistor for an output of 250 ± 5 VDC.

4. Protective Circuits

Close the control power switch(es) and verify that all the magnetic protective circuits are functioning properly. Use the elementary diagram to assure that all circuits are checked.

5. Motor and Booster Field Currents

a) Reconnect the windup, rider roll, slitter, and paper roll motor fields and check to see that the motor field currents are approximately correct and that any special field circuits such as field economy, field rheostats, etc., are operating properly. Refer to the machine data sheet of the elementary diagram for approximate field current values.

b) If the rider roll and slitter boosters or generators are being used, reconnect their fields and adjust the field resistors, to produce the desired booster or generator output voltage.

c) The above adjustments should be approximate only. Final adjustments will be made later.

6. Power Supply Module

Set up the power supply module as described in the Installation Adjustment section of the PS5 module instruction book GEK-7906. Calculate the tachometer voltage at rated machine speed and adjust the rheostat which is in parallel with the tachometer so that at this tachometer voltage + 100 V will exist from the slider to the full CW rheostat position, and then lock it. This 100 volts will then be used as the speed and CEMF signals.

7. Rate Reference Module

Set up the Rate Reference module as described in the Installation Adjustment section of the RRA module instruction book GEK-7905. Operate the run relay and contactor auxiliary relay interlocks by hand if necessary to permit checking the accel and decel times. Set the accel and decel times temporarily for 100 seconds.

8. Set up the VSA module in its "test" mode. That is, move the VSA test card from slot I to slot J. Test and set up the VSA as described in the Trouble Shooting section of the VSA instruction book GEK-7907. Note that the operation of the VSA module in its "test" mode permits manual control (by means of the test potentiometer RH4 in the VSA module) of the VSA output voltage. This feature can be used to check out the operation of the Silcomatic Power Supply. Return the VSA test card to its normal position.

9. Basic Regulator Checkout

Reconnect the windup motor armature leads. Disconnect the tachometer feedback to the VSA module and jumper the normally closed MX interlock so the voltage feedback remains in at all times. Set the Lead adjust pot RH6 at its midpoint and the Gain adjust pot RH7 at its maximum gain position (full CW). Start the drive and run it at a slow speed. Check to see that motor rotation is correct and tachometer generator polarity is correct (positive relative to common). Stop the drive and check that the speed relay CP drops out at less than one percent of rated machine speed. Remove the jumper from the MX interlock, reconnect the tachometer feedback to the VSA module, and reset the Gain adjust pot RH7 to its minimum gain position (full CCW).

10. Max Speed, Min Speed, and Jog

Initiate a windup run and gradually increase the speed. Adjust the Max Ref pot on the RRA module so that at full speed rotation of the handwheel controller the output of RRA is approximately -29 V. Next adjust the Max Speed pot in VSA for rated surface speed of the windup. Decrease the speed to minimum speed position on the handwheel and set the desired minimum speed by adjusting the Min Ref pot in the RRA module. Recheck and if necessary readjust the maximum speed. Initiate a jog operation and set the desired jog speed by adjusting the Jog Adjust pot RH3. Weaken the motor field a little if necessary to keep the Silcomatic voltage slightly below rated voltage when the drive is operating at rated speed.

11. Emergency Stop

Initiate an emergency stop and adjust the deceleration rate so that it is somewhat faster than the normal deceleration rate. Final adjustments of the normal and E-stop rates will be made later.

Speed regulator response (stability) will be set later.

12. Unwind Booster Fields and Tirastat Operation

Reconnect the unwind booster auxiliary field and check to see that there is approximately rated field current in that field. The booster voltage should be negative. Reconnect the booster main field and check to see that booster voltage is reduced. Reconnect the voltage feedback winding F11-F12 on the Tirastat and check to see that booster voltage is reduced to below three percent of its rated voltage. Reconnect the rest of the Tirastat windings.

13. Unwind Motor Field Current
(see also Step 22)

Reconnect the unwind motor field, rotate the MRH to the weak field position (normally this is full CW rotation of MRH), and check to see that the motor field current is reasonably close to the proper value. (This may have to be calculated using rated sheet speed, unwind core diameter, gear ratios, and unwind motor field data on the machine data sheet of the elementary diagram.) Now rotate the MRH to the full field position and check to see that the field current is reasonably close to the proper value.

14. Unwind Payout and Slack Take-Up

Reconnect the unwind contactor coil lead. Initiate an unwind slack take-up and adjust the STU ADJ pot on the unwind SCRX module to obtain the desired speed. Check to see that the unwind motor rotates in the proper direction. Initiate an unwind payout and adjust the PO ADJ pot to obtain the desired speed.

15. CEMF Regulator Bias

Temporarily jumper the R and ZSR interlocks in the output of CEMF Regulator Amplistat Circuit. Adjust the signal to Amplistat bias winding F9-F10 so that the Amplistat output is low enough to keep MRH from turning (or from continuing to turn once started), but high enough so that an extra five volts or so will cause MRH to turn. (Note: Because the Amplistat output voltage has considerable ripple, the sum of the positive and negative d-c voltmeter readings around the Amplistat, RS47, and MRH armature circuit may not appear to be zero.) Remove the jumper on the R and ZSR interlocks.

16. Current Regulator Adjust

Have a full roll of paper put in the unwind stand. Rotate the MRH to full CCW. Disconnect one of the MRH motor armature leads. Have the paper threaded through the machine and run the machine at a slow speed. Adjust the Max Current pot RH1 in the unwind SCRX module so that rated unwind current flows when the operator's Run tension adjust pot calls for maximum tension. Rotate the handwheel controller to the stall tension position and adjust the Stall Tension Adjust rheostat on the unwind SCRX to obtain the desired stall tension.

17. CEMF Regulator

Operate the machine at slow speed. Observe the Amplistat voltage and adjust the current compensation pot RH2 so that the Amplistat remains at a constant voltage plus or minus two volts as the unwind armature current is changed from zero to rated value. Increase and decrease the machine speed and adjust the CEMF adjust pot RH5 so that the Amplistat remains at a constant voltage plus or minus two volts as the speed changes from zero to rated speed. Reconnect the MRH motor armature leads and check to see that the MRH position changes as the unwind roll diameter decreases.

18. Inertia Compensation

Set the Accel and Decel adjust rheostats on the inertia compensation amplifier module ICA at their midposition. Accelerate and decelerate the machine and adjust the ICA gain pot to obtain approximately correct inertia compensation effect. If the inertia compensation effect appears to come into effect too soon, increase the ohms in the ICA lag adjust rheostat RH6; if it is too late, decrease the ohms in RH6. Make final adjustments on the Accel and Decel adjust pots if the amount of compensation required for accel and decel are different. (Note: It may be necessary to lengthen the deceleration time to keep the unwind current to a safe value when a large roll is on the unwind.)

19. Final Field Adjustments
(see also Step 22)

Make final adjustments to the windup motor field resistors so that the Silcomatic goes to slightly below rated voltage at rated machine speed. If the No. 2 Windup Drum has a load adjust rheostat, this rheostat should be approximately two-thirds shorted out during the adjustment. Adjust the resistor in series with the unwind motor field so that, when a full roll is on the unwind, the unwind motor CEMF is approximately equal to the Silcomatic rated volts at rated machine speed.

Make final adjustments to the rider roll motor field resistors so that when the machine is at rated speed (with the rider roll booster generator at minimum voltage and the rider roll not touching the paper), the rider roll surface speed is equal to machine rated speed. If two rider roll motors are used in a series connection, the field resistors should be adjusted so that approximately equal voltage is across each motor.

Make final adjustments to the rider roll booster field resistors so that the booster causes rated motor current to flow when the operator's load adjust rheostat is in its maximum load position (with the rider roll against the paper on the windup roll).

Make final adjustments to the slitter booster or generator fields to obtain the desired slitter speed at minimum winder speed (or desired slitter load if a mandrel for score slitting is used), and adjust the slitter motor fields to obtain the desired slitter speed at top sheet speed.

All of the above field resistor adjustments should be made when the motors and boosters are near their operating temperature. If the adjustments are made when the machines are cool, remember that the field resistance will increase (as much as 25 percent) as the field temperature increases from room temperature to machine rated temperature.

20. Final Accel and Decel Time Settings

Adjust the acceleration and deceleration times by adjusting the Accel and Decel pots on RRA to obtain as fast a normal accel and decel rate as desired, and decel for the desired E-stop rate without causing excessive (more than 200 percent of rated) Silcomatic, windup, or unwind current to flow. This operation should be checked for various roll diameter conditions on the wind and unwind stands.

21. Speed Regulator Stability

The speed of response of the speed regulator can be increased by reducing the value of capacitor C8 (in Fig. 3), or by reducing the resistance in the Gain rheostat RH7 (turn it CW). A reasonable response is desirable, but it is not necessary or desirable to make the system perform at the maximum speed of response possible. Do not remove so much capacitance that the drive is unstable. For a small step change in speed reference signal, if the drive overshoots the final value, increase the resistance of the Lead rheostat RH6 (turn it CW); if the drive speed creeps into the final value, reduce the resistance in the Lead reheostat RH6. Since the optimum setting will vary as the diameter of the roll of paper on the windup changes, a compromise setting is normally required.

22. For large unwind motors the MRH plate which is in series with small unwind motor fields is used as reference to an a-c/d-c SCR. Refer to the elementary diagram supplied with the equipment for tune-up adjustment.

23. From some sheet speed, select stall tension and adjust the switch volts adjust rheostat, as described under the Silcomatic speed regulator, so that the Silcomatic voltage does not change when the zero speed relay drops out.

## MAINTENANCE

### GENERAL

Periodic inspections and maintenance are important to continuous operation. Regular inspections should be performed so that replacements or repairs can be made before a failure occurs. Costly downtime can be reduced by following three basic rules:

1. Prepare a preventive-maintenance program consisting of items to be checked and recorded at each scheduled inspection. Sometimes frequent inspections are desirable on new equipments. The interval between inspections can then be increased at the user's discretion as he becomes more familiar with the equipment and particular operating conditions.

2. Keep on hand an adequate supply of renewal and repair parts.

3. Much time and trouble can be saved by buying renewal parts from the manufacturer who originally supplied the equipment. You are then certain such renewal parts will conform with the specifications for the original parts.

### MAINTENANCE

The maintenance schedule and activity outlined in publication GET-1195, "How to Maintain Industrial Control," apply to this control.

### RENEWAL PARTS

When ordering renewal parts, specify the complete catalog number of the part. If there is any doubt, include the panel catalog number and the device functional nomenclature.

## TROUBLE SHOOTING

During the design and manufacture of this control everything practical has been done to make it operate consistently and without failure. However, eventually some trouble may develop. To be prepared for this eventuality it is suggested that the user:

1. Keep the elementary diagram and instruction books handy.

2. Become familiar with the over-all concept of the drive and insofar as possible with the various circuits involved.

3. Keep portable instruments handy for checking voltage, current, and resistance. A portable industrial volt-ohm-milliameter with a 20,000 ohms-per-volt d-c sensitivity is the minimum recommended.

4. Keep on hand an adequate supply of spare parts.

Should the drive become inoperative or its performance change, care should be exercised during trouble shooting to prevent the drive from getting out of adjustment. Generally, the difficulty stems from a loose connection, broken wire, or failed component and it can be corrected without making additional adjustments to the control. Considerable time will be saved if the difficulty can be corrected without changing the component settings which would necessitate retuning the drive.

The first step in the trouble-shooting procedure should be to identify the problem with a specific area of the system, then systematically reduce this area to a specific module or component. The next step is to repair or replace the faulty component. The final step is to check the drive operation to be sure that no other secondary or contributing problems exist.

The following trouble-shooting guide may be of assistance in tracking down the problem to a general area of the control.

### TROUBLE-SHOOTING GUIDE

A. Drive does not start.

1. Coast stop relay CS not picked up or Emer Stop relay ES not picked up.

   Check protective interlocks in coil circuits. If unwind Tirastat undervoltage relay has not picked up, check the Tirastat operation (see GEI-80228).

2. Line contactor MW does not pick up.

   Check relay coil and interlock circuits.

3. Line contactor picks up but generator voltage remains at very low value.

   Check regulator reference circuit and rate reference unit RRA.

B. Drive starts but immediately goes out on overcurrent.

1. Check speed regulator feedback circuit (tach generator, etc.).

2. Check motor field circuits.

C. Windup operates but unwind does not hold tension.

1. Unwind will not operate in STU or PO mode.

   Check unwind contactor coil and relay circuits.

2. Unwind operates in STU and PO modes but does not regulate current.

   Check current reference circuits.

3. Unwind operates in STU and PO mode but goes out on overcurrent when switched to current mode.

   Check regulator current feedback circuit.

D. Unwind booster voltage gradually goes out of range.

1. Booster voltage changes with change in sheet speed.

   Check windup motor field excitation.

   Check MRH operation.

   Check Amplistat operation.

2. Booster voltage changes as roll diameter decreases.

   Check MRH operation.

   Check Amplistat operation.

Note: When checking the regulator module VSA, or the rate reference module RRA, or the operation of the speed relay CP, refer to the trouble-shooting procedures in the module instruction books for suggested steps and normal operating values.

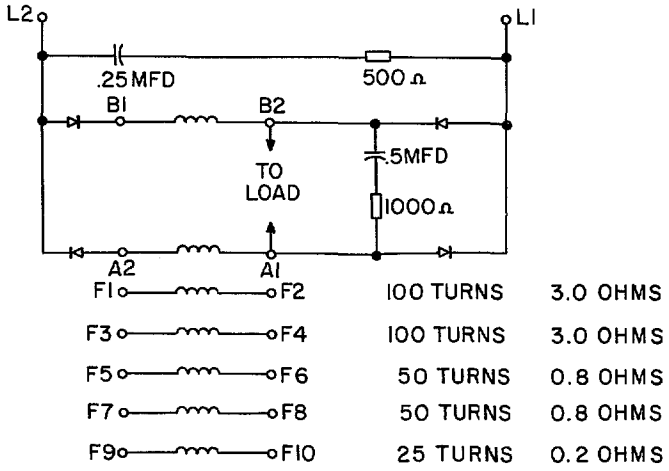
# APPENDIX

Amplistat Data Cat. #9T95424

Input: 120 v 60 cy 1 ph

Output: Rated load 1. 0 amps

Internal circuit:




Characteristic

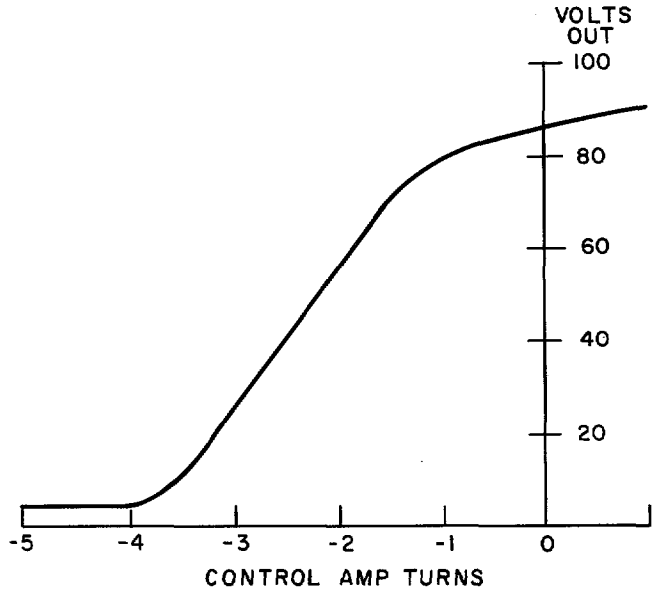



*Fig. A1. 75 VA Amplistat*

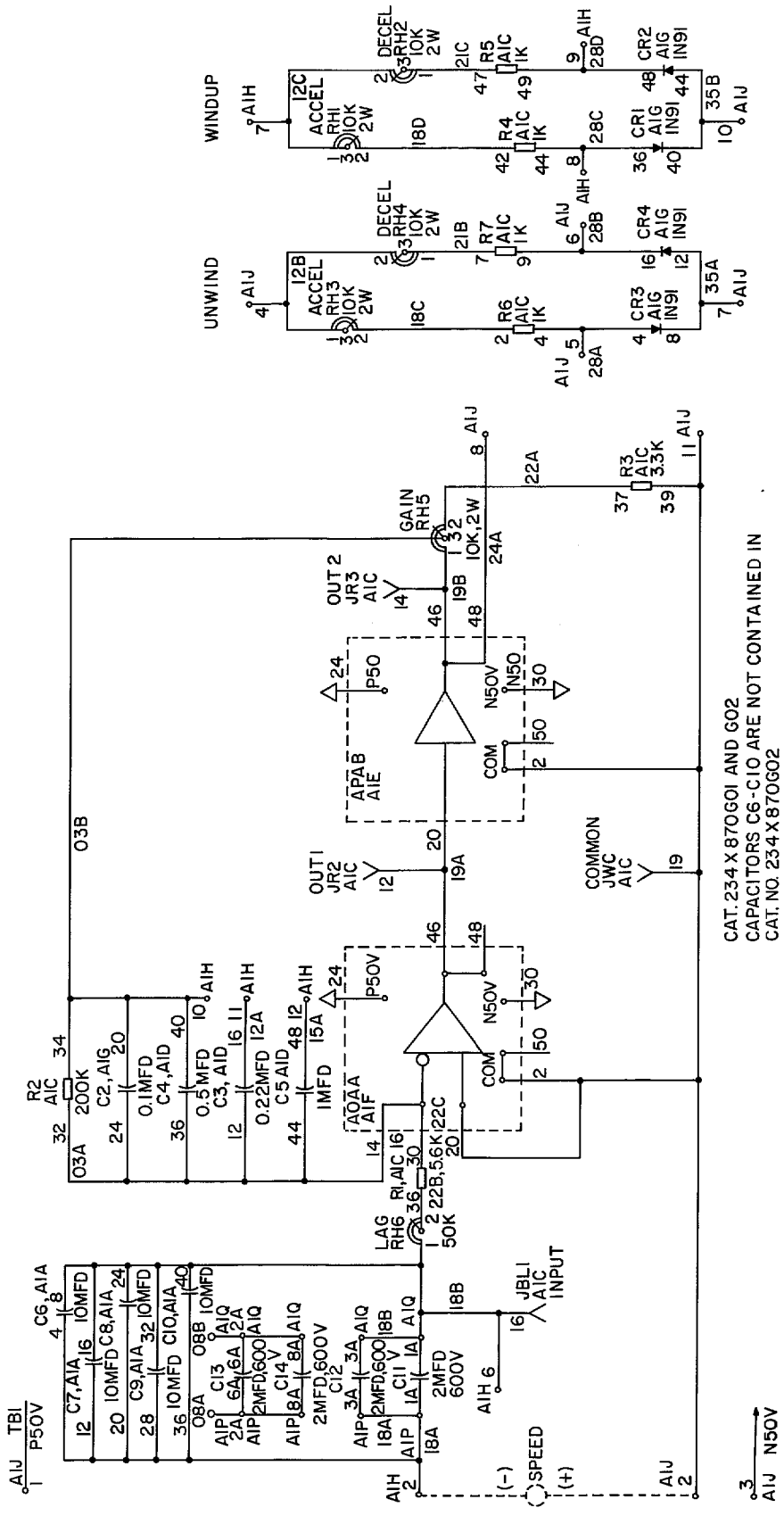



*Fig. A2. Inertia compensation amplifier (ICA)*

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